

Cwc15 stabilizes the spliceosomal catalytic interactions

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1. Abstract

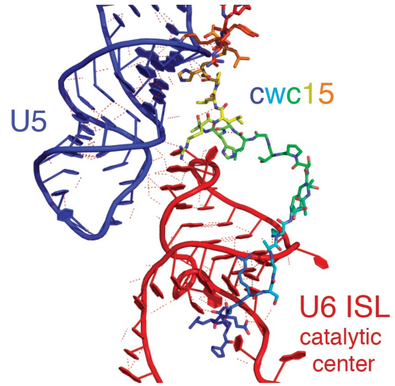
To better understand structural rearrangements between the catalytic steps of splicing, we performed a structural search of the cryo-EM spliceosomal structures to identify proteins in the close proximity to the catalytic center, likely to affect splicing catalysis.

We focused on a poorly characterized protein, Cwc15, which through its disordered N-terminal domain interacts directly with U6 snRNA at the catalytic center. To explain the role of Cwc15 in splicing, we analyzed yeast *S. cerevisiae* strains carrying Prp8, U6 and U2 alleles as markers for spliceosomal rearrangements at various splicing stages.

In the *cwc15Δ* strain, splicing of introns with mutations limiting the 2nd step (BS-c, gAG, A3c) is inhibited, whereas those limiting for the 1st step (BS-g, G5a, A3u, BS-U257c) is not affected, suggesting that interactions of Cwc15 stabilize the 2nd step catalytic conformation.

Analysis of the *cwc15Δ* strain in the context of the first- and second-step alleles of Prp8 revealed a strong exacerbation of the first step allele defects. This implies that wt Cwc15 improves such a transition from the first to second steps, improving the 2nd step of splicing. This is consistent with the observed inhibition of splicing for intron mutants limiting for the 2nd step by deletions and point mutants in Cwc15.

2. Cwc15 interacts directly with U5 and U6 snRNA

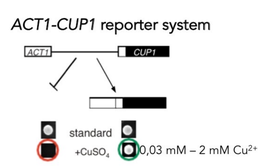


We performed a structural search of the cryo-EM spliceosome C-complex structure (Galej, 2016) to select proteins in close proximity to the catalytic center of the spliceosome, suggesting their modulating role in splicing. The list was then narrowed to proteins whose deletions change stress resistance, based on the data provided by (Brown, 2006).

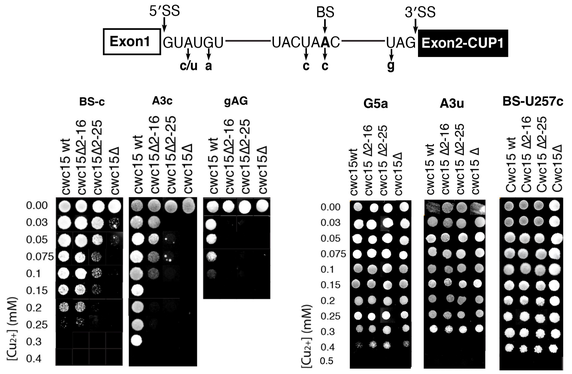
We focused on an uncharacterized protein, Cwc15 which interacts with its disordered domain directly with U6 RNA at the catalytic center of the spliceosome and its deletion increases oxidative stress resistance (Brown, 2006).

The Cwc15 protein (rainbow) interacts with the U6 RNA (red) and U5 RNA (blue) as observed in the cryo-EM spliceosome structure (PDB ID: 5LJ3; Galej, 2016).

3. *Cwc15Δ* inhibits splicing of introns with mutations limiting the 2nd step



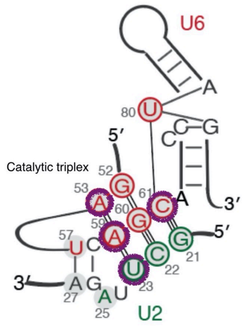
In the *cwc15Δ* strain, splicing of various reporters of *ACT1-CUP1* system with mutations limiting the 2nd step is inhibited (left panel), whereas those limiting for the 1st step is not affected (right panel). This result suggests that interactions of wt Cwc15 stabilize the 2nd step catalytic conformation.



BS-c - limiting for the I & II step
A3c - limiting for the II step, U6 hyperstabilization/limiting for transition
gAG - limiting for the II step

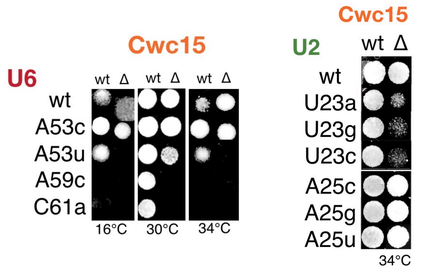
G5a - limiting for the I step, defective in base pairing to both U1 and U6 snRNAs
A3u - limiting for the I step
BS-U257c - limiting for the I step

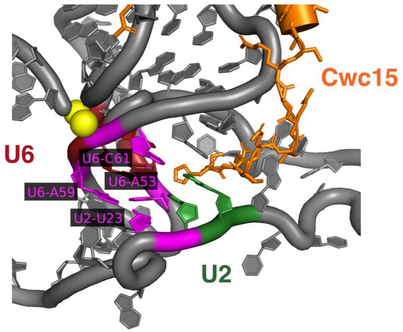
4. Cwc15 interacts with the catalytic triplex



The catalytic site of the spliceosome as a diagram and a structural model (PDB ID: 6EXN; Wilkinson, 2017). The positions affected by the Cwc15 deletion are indicated in the magenta.

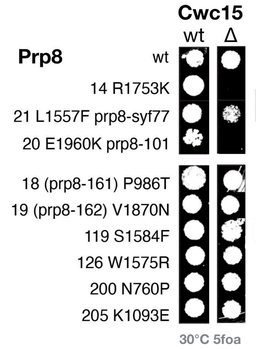
The deletion of cwc15 is synthetically lethal with U6-A59c and -C61a alleles, suggesting that Cwc15 interacts with the catalytic triplex. This is further supported by the observed growth defects at 34°C of the *Δcwc15 strain* with U2-U23a/g/c (again, the catalytic triplex residue).





5. *ΔCwc15* strain revealed exacerbation of the first-step alleles of Prp8

cwc15Δ strain in the context of the first- and second-step alleles of Prp8 (Liu, 2007) revealed an exacerbation of the first step allele defects (top panel), and no effect with the second-step alleles (bottom panel). This implies that wt Cwc15 improves such a transition from the first to second steps, improving the 2nd step of splicing.



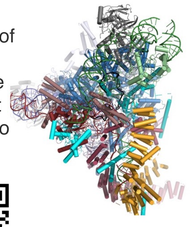
first-step alleles suppress first-step defects due to intron mutations while exacerbating second-step defects

second-step alleles suppress a large number of intron mutations at 5′SS, branch site, and 3′SS by improving the second step while also reducing the efficiency of the first step

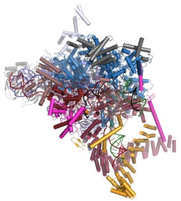
6. Summary

- The N-terminal disordered domain of Cwc15 interacts with U6 RNA at the catalytic center of the spliceosome.
- The results suggest that Cwc15 stabilizes the 2nd step catalytic conformation
- This is still a work in progress: detailed analysis of Cwc15 with point mutations is being performed right now and the mechanism is being explored.

Visit **rna-tools,** a toolbox to analyze sequences, structures and simulations of RNA (and more) and **PyMOL4Spliceosome**, all spliceosome structures in one PyMOL session to get access to exciting bioinformatics tools to deepen your structural spliceosomal analyses.



yC_5LJ3.pse



yCs_5MPS.pse

GitHub

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